Graphic Era HILL UNIVERSITY

Mid Term (Odd) Semester Examination October 2024

Roll no.....

Name of the Course and semester: **B.Tech. CE, Semester III**
Name of the Paper: **Mathematics-III**
Paper Code: **TMA 302**
Time: 1.5 hour

Maximum Marks: 50

Note:
(i) Answer all the questions by choosing any one of the sub questions
(ii) Each question carries 10 marks.

Q1. (10 Marks)

a. Express the function $f(x) = \begin{cases} 1 & if\ |x| \le 1 \\ 0 & if\ |x| > 1 \end{cases}$ as a Fourier integral hence evaluate

$$\int_0^\infty \frac{\sin\lambda \cos\lambda x}{\lambda} d\lambda$$ **(CO1)**

OR

b. Find the Fourier transform of $f(x) = \begin{cases} 1 + \frac{x}{a} & if\ -a < x < 0 \\ 1 - \frac{x}{a} & if\ \ 0 < x < a \end{cases}$ **(CO1)**

Q2. (10 Marks)

a. Find the Fourier sine and cosine transform of e^{-x}. **(CO1)**

OR

b. Find the Fourier sine transform of e^{-ka}, x is greater than or equal to zero and k is a positive constant. **(CO1)**

Q3. (10 Marks)

a. If $f(z) = u + iv$ is analytic function and $u - v = e^x(\cos y - \sin y)$, find $f(z)$ in terms of z. **(CO2)**

OR

b. If $u = x^2 - y^2, v = -\frac{y}{(x^2+y^2)}$, then show that both u and v is harmonic, but $u + iv$ is not an analytic function of z. **(CO2)**

Q4. (10 Marks)

a. State Cauchy's integral theorem. Evaluate $\oint \frac{z}{(9-z^2)(z+i)} dz$, where C is the circle $|z| = 2$ by using Cauchy's integral formula **(CO2)**

Mid Term (Odd) Semester Examination October 2024

OR

b. Find the analytic function $f(z) = u(x, y) + iv(x, y)$ of which the real part $u(x, y) = e^x(x \cos y - y \sin y)$. **(CO2)**

Q5. (10 Marks)

a. Show that the function $u(x, y) = \frac{1}{2}\log(x^2 + y^2)$ is harmonic then find the analytic function $f(z) = u + iv$. **(CO2)**

OR

b. If $f(z) = u + iv$ is analytic in a domain D, then u, v satisfy the equations $\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}$, $\frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$. **(CO2)**